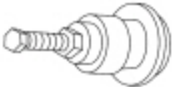


# Cylinder Head



## Special Tool(s)

|                                                                                                |                                                                    |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <br>ST2806-A   | Alignment Pins, Cylinder Head<br>303-1040 (SR-015486)              |
| <br>ST3245-A   | Installer, Crankshaft Damper and Crankshaft Front Seal<br>303-1508 |
| <br>ST1328-A   | Installer, Front Cover Oil Seal<br>303-335 (T88T-6701-A)           |
| <br>ST1377-A | Lifting Bracket, Engine<br>303-F047 (014-00073) or equivalent      |
| <br>ST1438-A | Strap Wrench<br>303-D055 (D85L-6000-A) or equivalent               |
| <br>ST3110-A | Support Brackets, Engine<br>303-1507 or equivalent                 |

## Material

| Item                                                                                                         | Specification    |
|--------------------------------------------------------------------------------------------------------------|------------------|
| Motorcraft® Metal Surface Prep Wipes<br>ZC-31-B                                                              | —                |
| Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) | WSS-<br>M2C945-A |

|                                               |               |
|-----------------------------------------------|---------------|
| XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)     |               |
| Motorcraft® Silicone Gasket and Sealant TA-30 | WSE-M4G323-A4 |
| Motorcraft® Silicone Gasket Remover ZC-30-A   | —             |
| Motorcraft® Threadlock 262 TA-26              | WSK-M2G351-A6 |

### All cylinder heads

1. Position the crankshaft keyway at the 11 o'clock position.
2. **NOTICE: Make sure all coolant residue and foreign material are cleaned from the block surface and cylinder bore. Failure to follow this instruction may result in engine damage.**

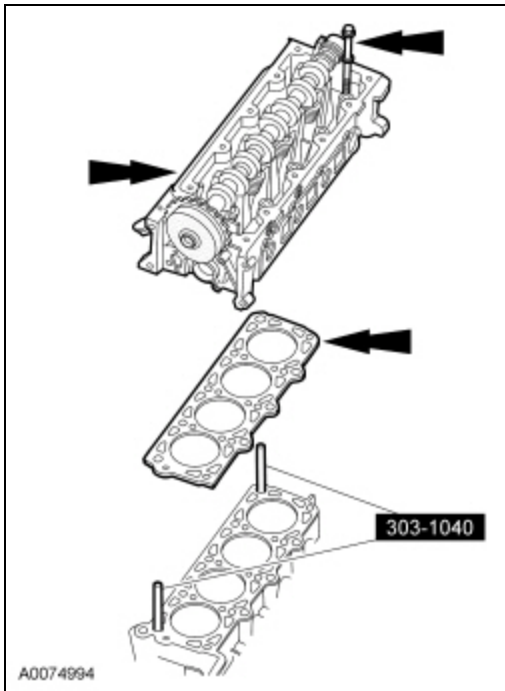
**NOTE:** *The use of sealing aids (aviation cement, copper spray and glue) is not permitted. The gasket must be installed dry.*

**NOTE:** *The cylinder head bolts must be discarded and new bolts installed. They are a tighten-to-yield design and cannot be reused.*

**NOTE:** *Do not turn the crankshaft until instructed to do so.*

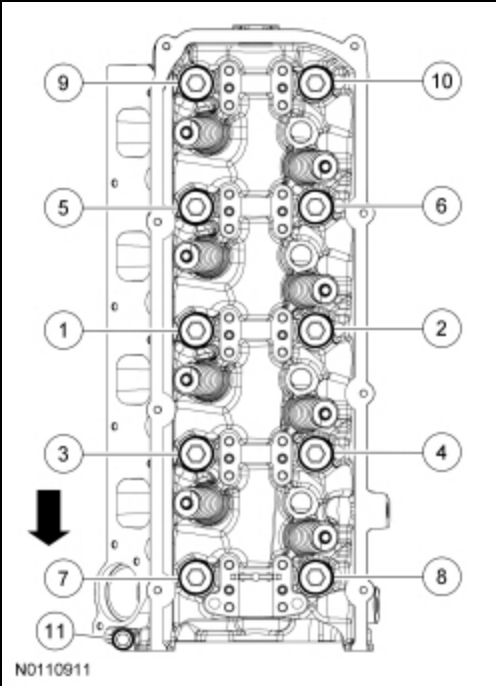
**NOTE:** *LH shown, RH similar.*

Using the Cylinder Head Alignment Pins, position the cylinder head gaskets and cylinder heads over the dowels and install the 20 cylinder head bolts loosely.

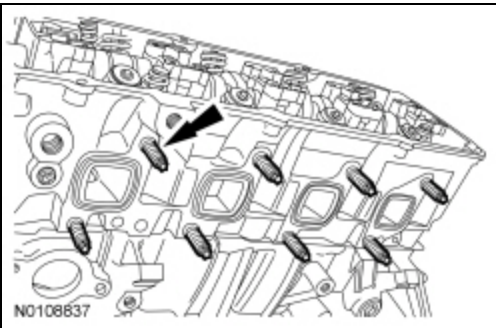


### LH cylinder head

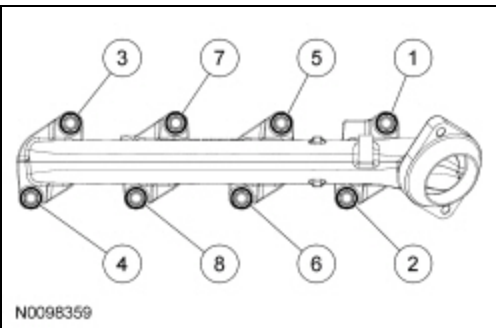
3. Tighten the 11 bolts in 6 stages, in the sequence shown.
  - Stage 1: Tighten the M12 fasteners to 25 Nm (18 lb-ft).
  - Stage 2: Tighten the M12 fasteners to 60 Nm (44 lb-ft).
  - Stage 3: Tighten the M12 fasteners an additional 90 degrees.
  - Stage 4: Tighten the M12 fasteners an additional 90 degrees.
  - Stage 5: Tighten the M8 fastener to 20 Nm (177 lb-in).
  - Stage 6: Tighten the M8 fastener an additional 45 degrees.



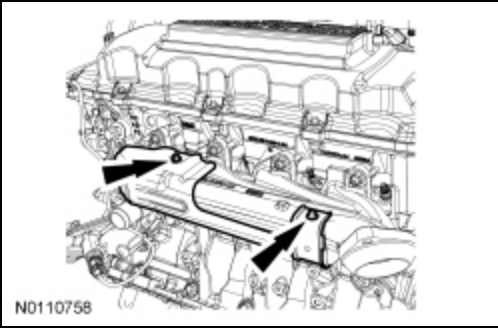
4. Install 8 new exhaust manifold-to-cylinder head studs.
- Tighten to 12 Nm (106 lb-in).



5. Using new exhaust manifold nuts, position the LH exhaust manifold and install the 8 nuts in 2 stages in the sequence shown.
- Stage 1: Tighten to 25 Nm (18 lb-ft).
  - Stage 2: Tighten to 32 Nm (24 lb-ft).

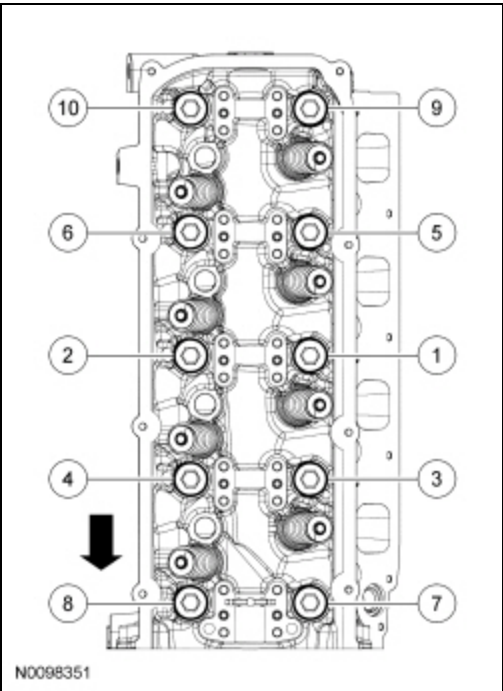


6. Position the LH exhaust manifold heat shield and install the 2 bolts.
- Tighten to 12 Nm (106 lb-in).

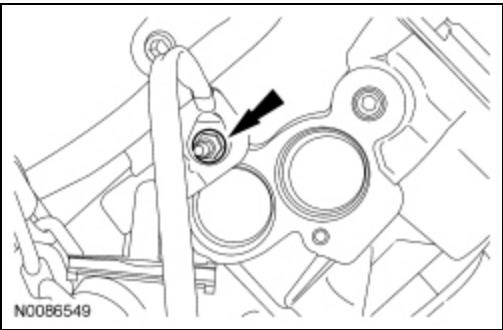


## RH cylinder head

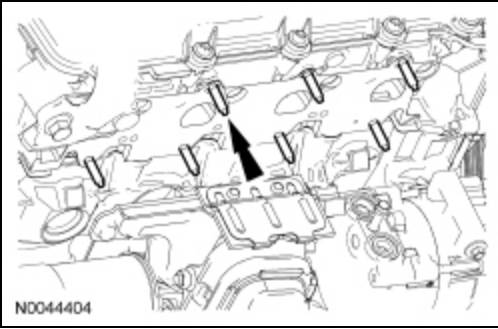
7. Tighten the 10 bolts in 4 stages, in the sequence shown.
  - Stage 1: Tighten the M12 fasteners to 25 Nm (18 lb-ft).
  - Stage 2: Tighten the M12 fasteners to 60 Nm (44 lb-ft).
  - Stage 3: Tighten the M12 fasteners an additional 90 degrees.
  - Stage 4: Tighten the M12 fasteners an additional 90 degrees.



8. Position the ground strap and install the nut.
  - Tighten to 10 Nm (89 lb-in).

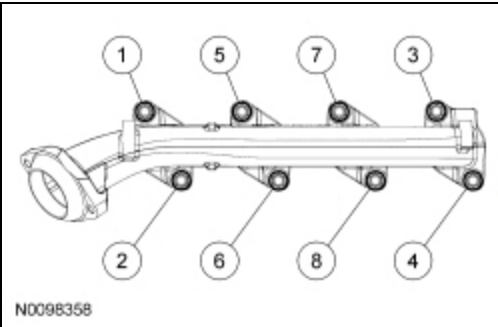


9. Install 8 new exhaust manifold-to-cylinder head studs.
  - Tighten to 12 Nm (106 lb-in).



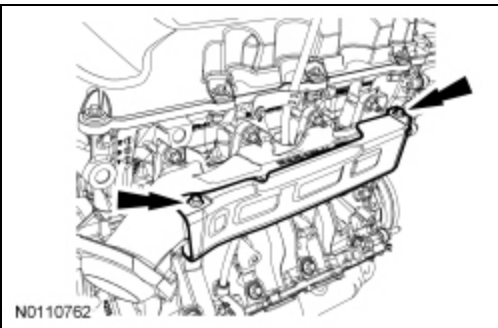
10. Using new exhaust manifold nuts, position the LH exhaust manifold and install the 8 nuts in 2 stages in the sequence shown.

- Stage 1: Tighten to 25 Nm (18 lb-ft).
- Stage 2: Tighten to 32 Nm (24 lb-ft).



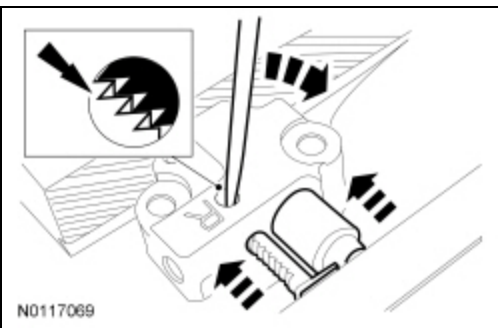
11. Position the RH exhaust manifold heat shield and install the 2 bolts.

- Tighten to 12 Nm (106 lb-in).

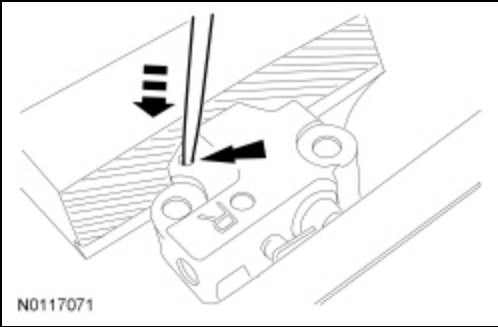


## All cylinder heads

12. Using a small pick, carefully push the tensioner rack pawl retainer away from the rack pawl and compress the tensioner plunger and rack using a vise.

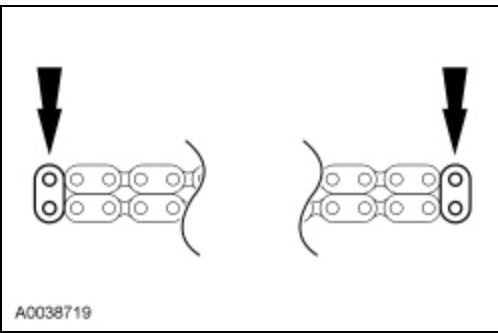


13. Install a small pick into the tensioner to hold the rack pawl and plunger in the seated position for tensioner installation.



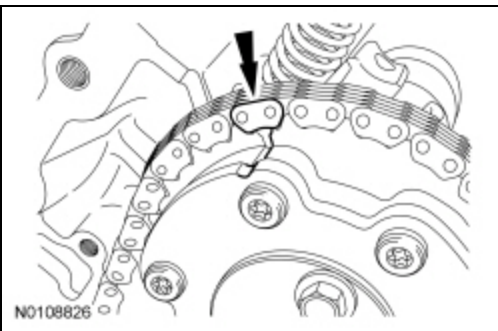
14. Remove the tensioner from the vise.

15. If the blue links are not visible, mark 2 links on each end, and use as timing marks.

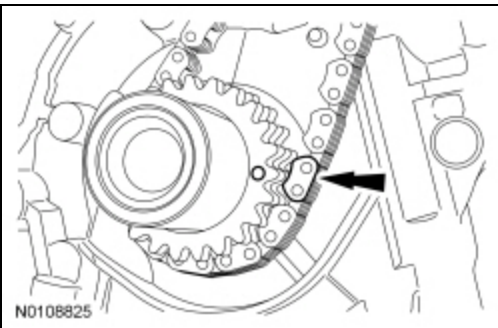


16. **NOTE:** Make sure the upper half of the timing chain is below the tensioner arm dowel and above the chain guide pin.

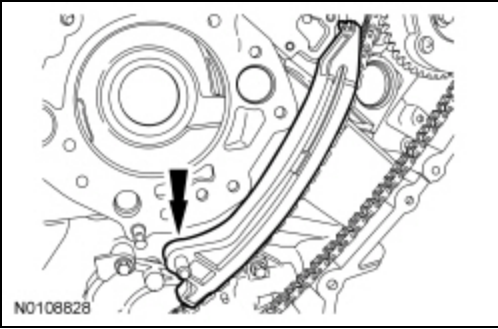
Position the upper end of the LH (inner) timing chain on the LH camshaft phaser and sprocket, aligning the timing mark on the outer flange of the camshaft phaser and sprocket with the single blue (marked) link on the chain.



17. Position the lower end of the LH (inner) timing chain on the crankshaft sprocket, aligning the timing mark on the outer flange of the crankshaft sprocket with the single blue (marked) link on the chain and install the crankshaft sprocket onto the crankshaft and verify the crankshaft keyway is at the 11 o'clock position.

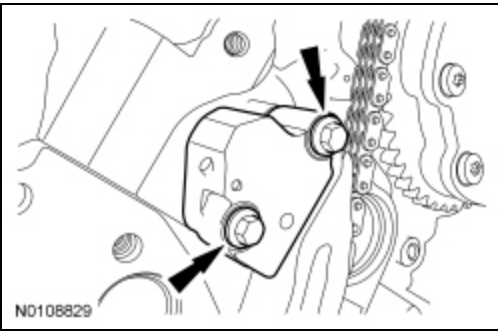


18. Install the LH timing chain tensioner arm.



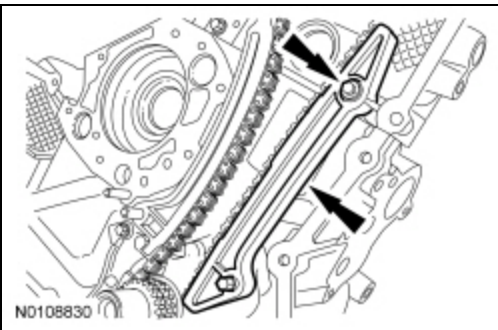
19. Position the LH timing chain tensioner and install the 2 bolts in 2 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.



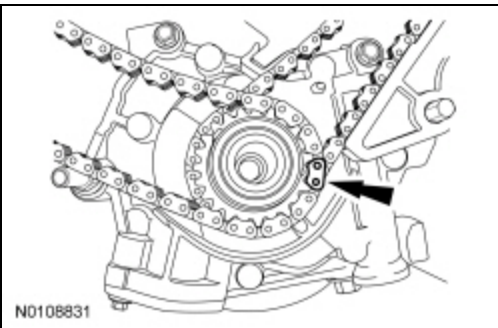
20. Position the LH timing chain guide and install the bolt in 2 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.



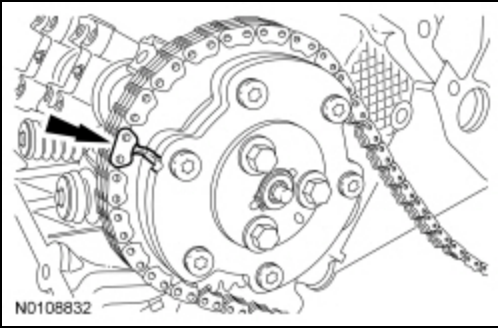
21. Remove the retaining clip from the LH timing chain tensioner.

22. Position the lower end of the RH (outer) timing chain on the crankshaft sprocket, aligning the timing mark on the sprocket with the single blue (marked) chain link.



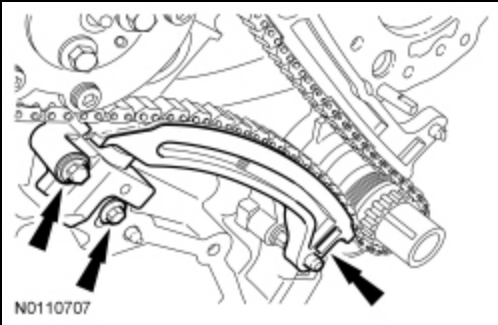
23. Position the upper end of the RH (outer) timing chain on the RH camshaft phaser and sprocket, aligning the timing mark on the outer flange of the camshaft phaser and sprocket with the single blue (marked) link on the chain.





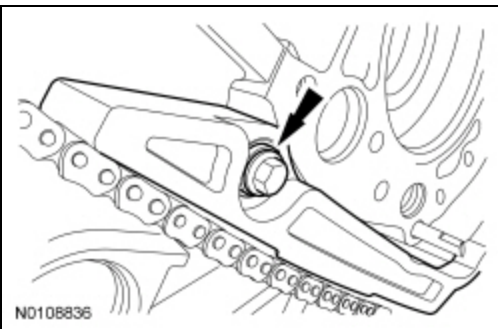
24. Position the RH timing chain tensioner arm on the dowel pin and the RH timing chain tensioner and install the 2 bolts in 2 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.



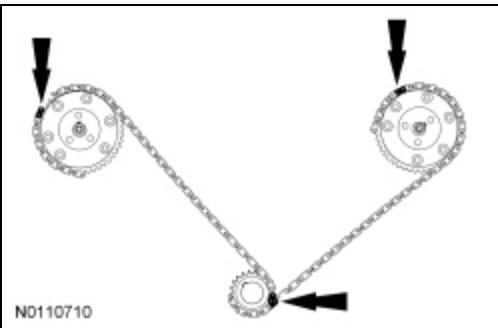
25. Position the RH timing chain guide and install the bolt in 2 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.



26. Remove the retaining clip from the RH timing chain tensioner.

27. As a post-check, verify correct alignment of all timing marks.



28. Lubricate the rocker arm shaft-to-valve tip and rocker arm-to-camshaft lobe contact area with clean engine oil prior to installation.

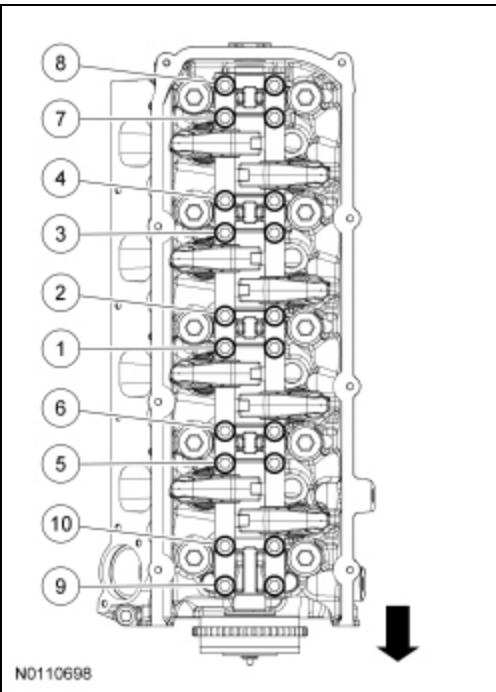
29. Rotate the engine clockwise until the No. 1 cylinder camshaft intake lobe is positioned so that the intake valve would be fully opened.



30. Position the LH intake rocker arm shaft assembly and install the bolts finger-tight.

31. Tighten the 10 LH intake rocker arm shaft assembly bolts in the sequence shown, in 3 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten to 20 Nm (177 lb-in).
- Stage 3: Tighten an additional 60 degrees.

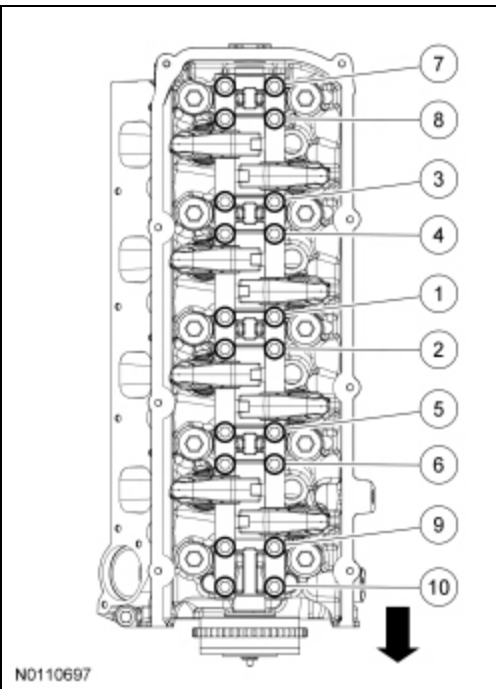


32. Rotate the engine clockwise until the No. 1 cylinder camshaft exhaust lobe is positioned so that the exhaust valve would be fully opened.

33. Position the LH exhaust rocker arm shaft assembly and install the bolts finger-tight.

34. Tighten the 10 LH exhaust rocker arm shaft assembly bolts in the sequence shown, in 3 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten to 20 Nm (177 lb-in).
- Stage 3: Tighten an additional 60 degrees.

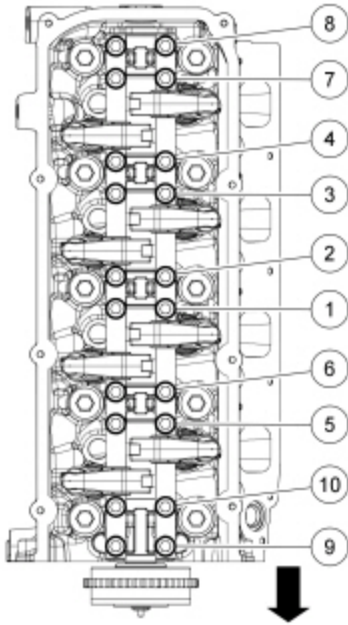


35. Rotate the engine clockwise until the No. 1 cylinder camshaft intake lobe is positioned so that the intake valve would be fully closed.

36. Position the RH intake rocker arm shaft assembly and install the bolts finger-tight.

37. Tighten the 10 RH intake rocker arm shaft assembly bolts in the sequence shown, in 3 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten to 20 Nm (177 lb-in).
- Stage 3: Tighten an additional 60 degrees.

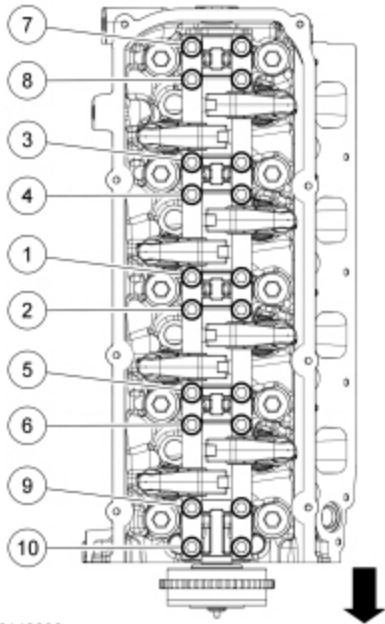


38. Rotate the engine clockwise until the No. 1 cylinder camshaft exhaust lobe is positioned so that the exhaust valve would be fully closed.

39. Position the RH exhaust rocker arm shaft assembly and install the bolts finger-tight.

40. Tighten the 10 RH exhaust rocker arm shaft assembly bolts in the sequence shown, in 3 stages.

- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten to 20 Nm (177 lb-in).
- Stage 3: Tighten an additional 60 degrees.



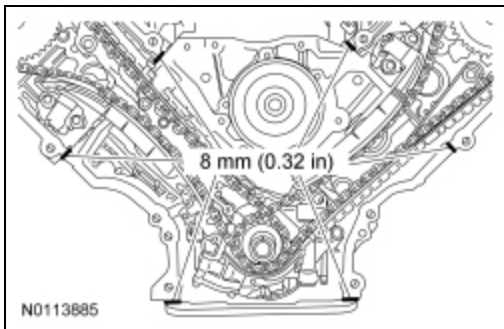
41. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to

**remove all traces of old sealant.**

**NOTE:** *If the engine front cover is not secured within 6 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

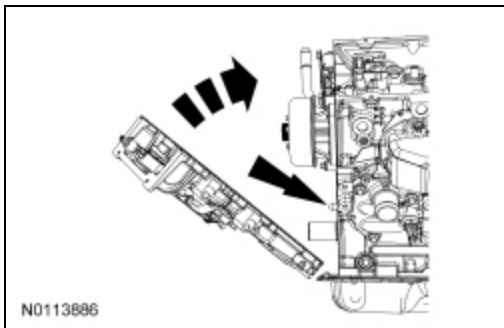
**NOTE:** *Make sure that the engine front cover gasket is in place on the engine front cover before installation.*

Apply a bead of silicone gasket and sealant along the cylinder head-to-cylinder block surface and the oil pan-to-cylinder block surface, at the locations shown.



42. **NOTICE:** The Variable Camshaft Timing (VCT) variable force solenoid pins must be fully depressed to avoid interference with the VCT valve tips when installing the engine front cover. Failure to follow these instructions can result in damage to the engine.

Install new engine front cover gaskets on the engine front cover. With the VCT variable force solenoid pins fully depressed, position the engine front cover onto the dowels. Install the fasteners finger-tight.



43. Tighten the 20 engine front cover fasteners in the sequence shown in 2 stages.

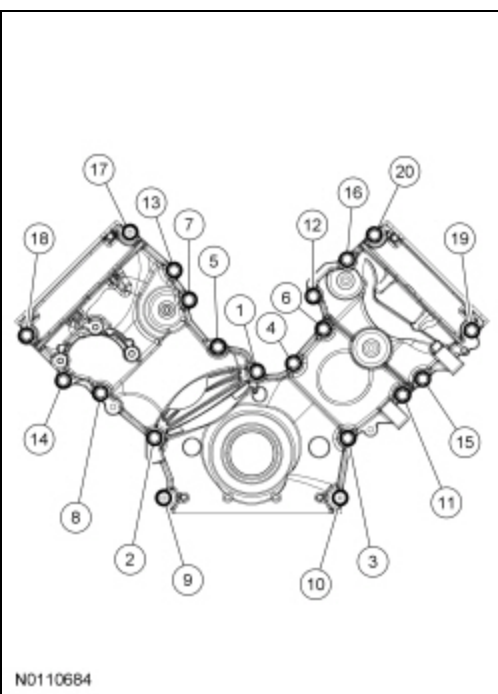
Stage 1: Tighten all fasteners to 10 Nm (89 lb-in).

Stage 2: Tighten all fasteners to 20 Nm (177 lb-in).

Stage 3: Tighten all fasteners an additional 45 degrees.

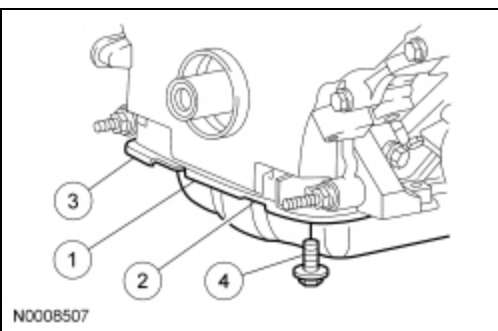
| Item | Part Number | Description                                 |
|------|-------------|---------------------------------------------|
| 1    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 2    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 3    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 4    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 5    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 6    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 7    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 8    | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |

| Item | Part Number | Description                                 |
|------|-------------|---------------------------------------------|
| 9    | W713462     | Stud, Hex Head Pilot, M8 x 33 + M10 x 30    |
| 10   | W713462     | Stud, Hex Head Pilot, M8 x 33 + M10 x 30    |
| 11   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 12   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 13   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 14   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 15   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 16   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 17   | W713461     | Stud, Hex Head Pilot, M8 x 33 + M8 x 27     |
| 18   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 19   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |
| 20   | W713261     | Bolt, Hex Flange Head Pilot, M8 x 1.25 x 33 |

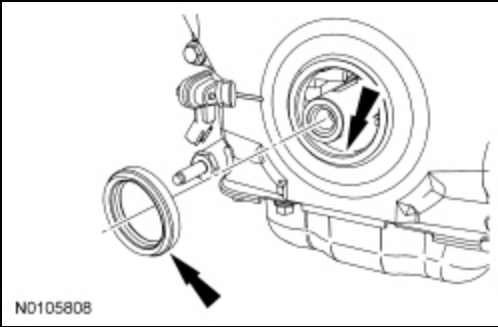


44. Loosely install the 4 bolts, then tighten in 2 stages, in the sequence shown.

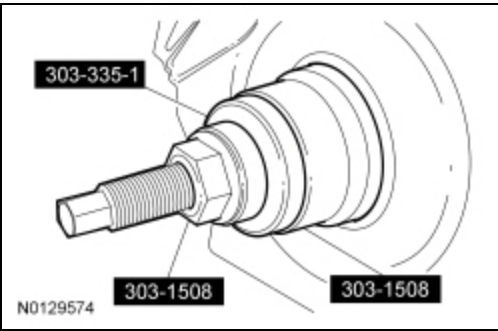
- Stage 1: Tighten to 10 Nm (89 lb-in).
- Stage 2: Tighten an additional 45 degrees.



45. Lubricate the engine front cover and the crankshaft seal inner lip with clean engine oil.

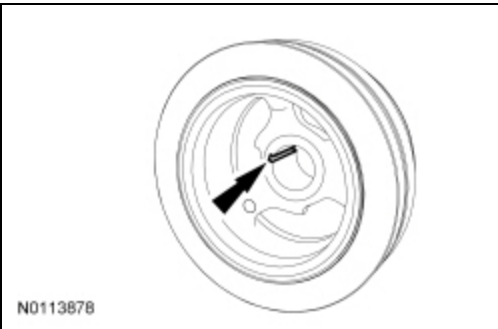


46. Using the Crankshaft Damper and Crankshaft Front Seal Installer and the Front Cover Oil Seal Installer, install a new crankshaft front seal into the engine front cover.

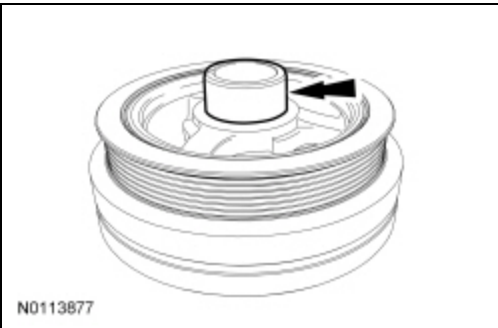


47. **NOTE:** *If not secured within 6 minutes, the sealant must be removed and the sealing area cleaned with metal surface prep and silicone gasket remover. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

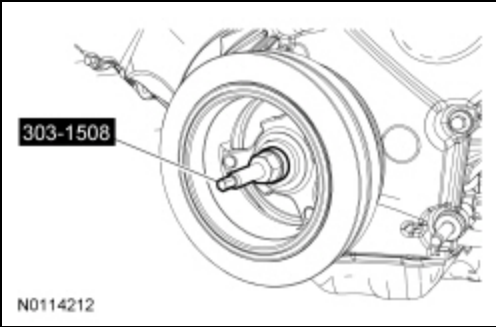
Apply silicone gasket and sealant to the Woodruff key slot in the crankshaft pulley.



48. Lubricate the crankshaft pulley sealing area with clean engine oil prior to installation.

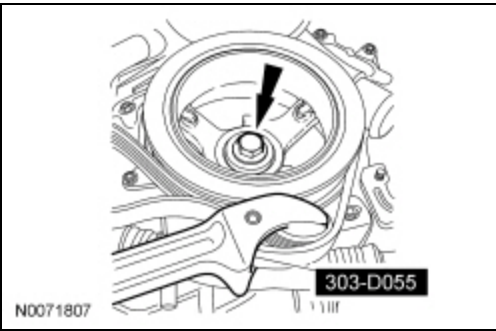


49. Using the Crankshaft Vibration Damper Installer, install the crankshaft pulley.



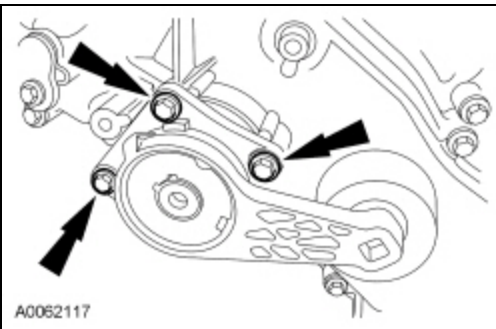
50. Using the Strap Wrench, install a new crankshaft pulley bolt and the original washer, tighten the bolt in 2 stages.

- Stage 1: Tighten to 175 Nm (129 lb-ft).
- Stage 2: Tighten an additional 90 degrees.



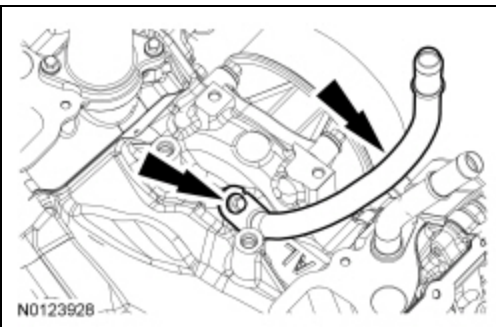
51. Install the accessory drive belt tensioner and the 3 bolts.

- Tighten to 25 Nm (18 lb-ft).



52. Install the heater coolant tube and the bolt.

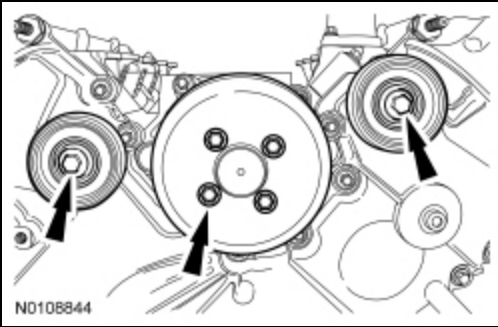
- Tighten to 10 Nm (89 lb-in).



53. Install the 2 accessory drive idler pulleys, the coolant pump pulley and the 6 bolts.

- Tighten to 25 Nm (18 lb-ft).



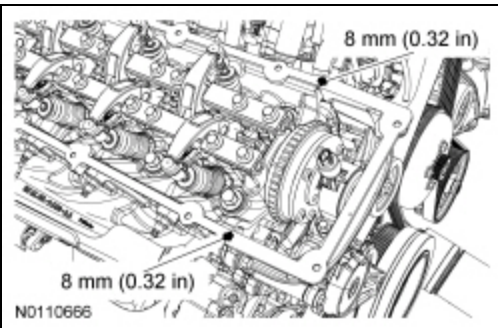


**54. NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

Clean the valve cover mating surface with silicone gasket remover and metal surface prep. Follow the directions on the packaging.

**55. NOTE:** *If not secured within 6 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

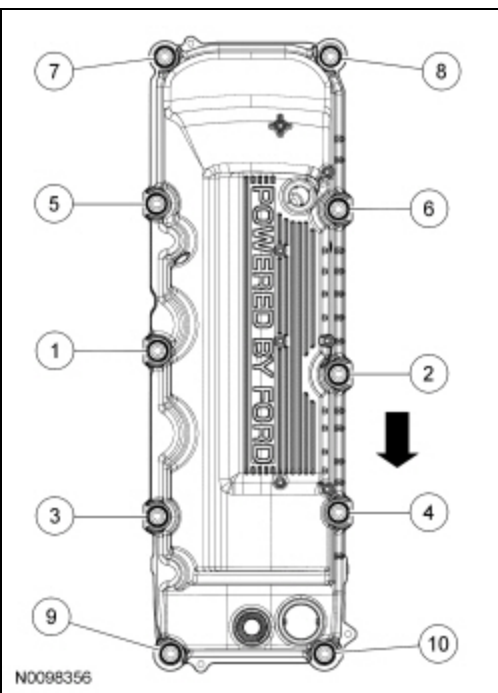
Apply silicone gasket and sealant in 2 places where the engine front cover meets the cylinder head.



**56. NOTICE:** Install the valve cover carefully, or the Variable Camshaft Timing (VCT) variable force solenoid may be damaged.

Position the RH valve cover and new gasket on the cylinder head and tighten the 10 fasteners in the sequence shown.

- Tighten to 10 Nm (89 lb-in).



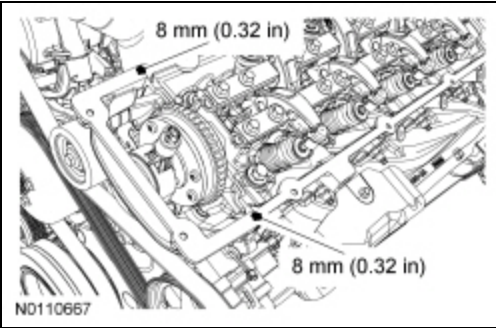


**57. NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

Clean the valve cover mating surface with silicone gasket remover and metal surface prep. Follow the directions on the packaging.

**58. NOTE:** *If not secured within 6 minutes, the sealant must be removed and the sealing area cleaned. To clean the sealing area, use silicone gasket remover and metal surface prep. Follow the directions on the packaging. Failure to follow this procedure can cause future oil leakage.*

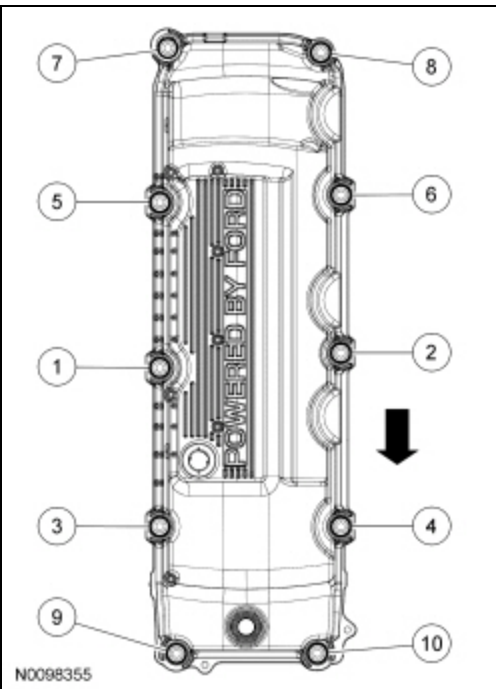
Apply silicone gasket and sealant in 2 places where the engine front cover meets the cylinder head.



**59. NOTICE:** Install the valve cover carefully, or the Variable Camshaft Timing (VCT) variable force solenoid may be damaged.

Position the LH valve cover and new gasket on the cylinder head and tighten the 10 fasteners in the sequence shown.

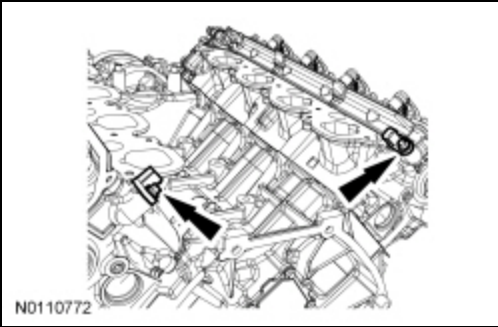
- Tighten to 10 Nm (89 lb-in).



**60. NOTE:** Lubricate the O-ring seals with clean engine oil prior to installation.

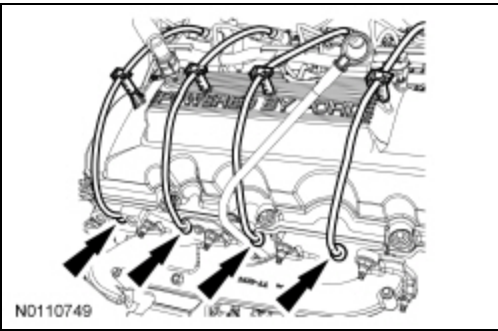
Install the LH and RH Camshaft Position (CMP) sensors and the 2 bolts.

- Tighten to 10 Nm (89 lb-in).



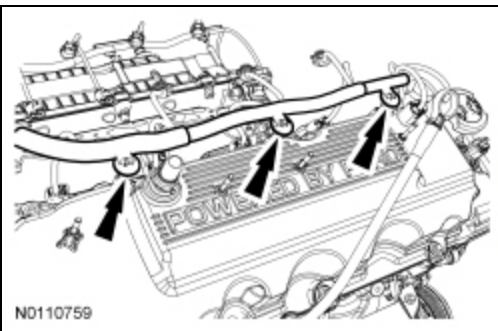
61. **NOTE:** *RH shown, LH similar.*

Connect the 8 ignition wires to the valve cover retainers and the lower spark plugs.

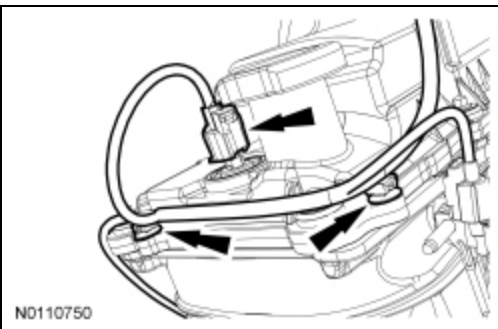


62. Position the wiring harness onto the engine assembly.

63. Connect the 3 engine wiring harness position retainers to the RH valve cover.

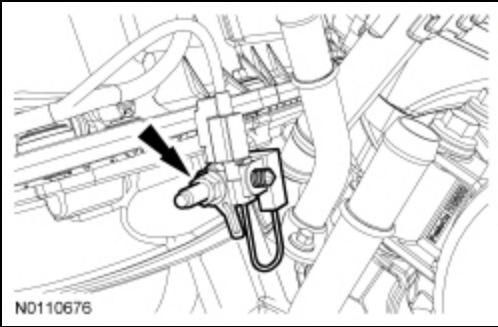


64. Connect the RH VCT variable force solenoid electrical connector and the 2 wiring harness retainers.



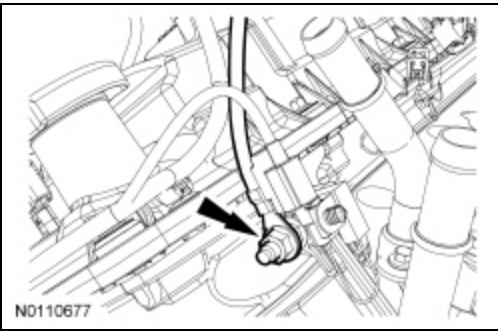
65. Position the radio ignition interference capacitor and install the nut.

- Tighten to 25 Nm (18 lb-ft).



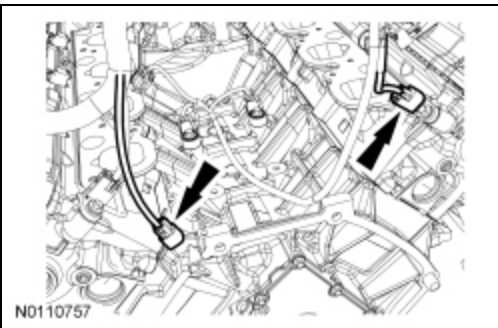
66. Position the ground strap and install the nut.

- Tighten to 25 Nm (18 lb-ft).

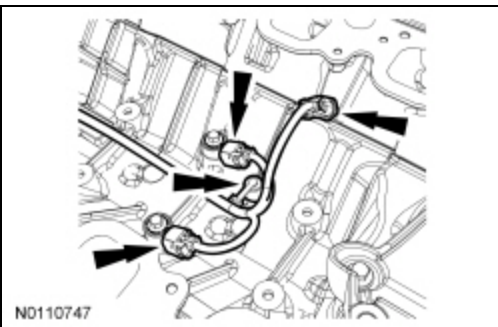


67. **NOTE:** *Lubricate the O-ring seals with clean engine oil prior to installation.*

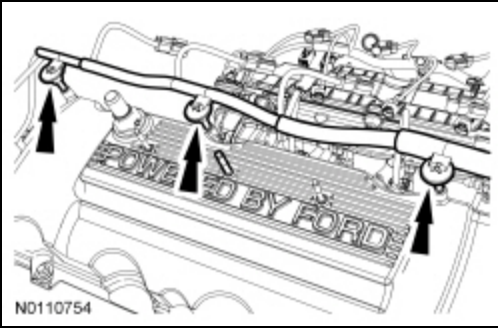
Connect the LH and RH CMP sensor electrical connectors.



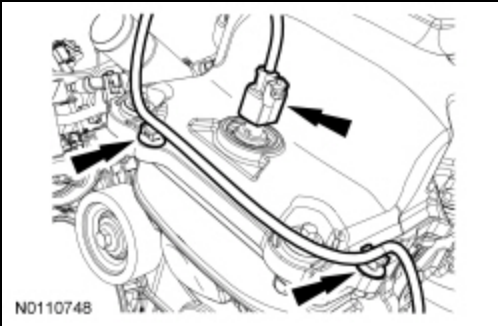
68. Connect the wiring harness retainer and the Knock Sensor (KS) and Cylinder Head Temperature (CHT) sensor electrical connectors.



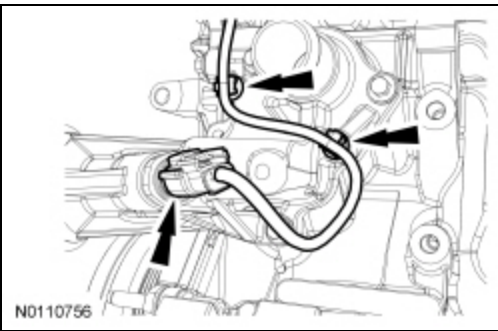
69. Connect the 3 engine wiring harness position retainers to the LH valve cover.



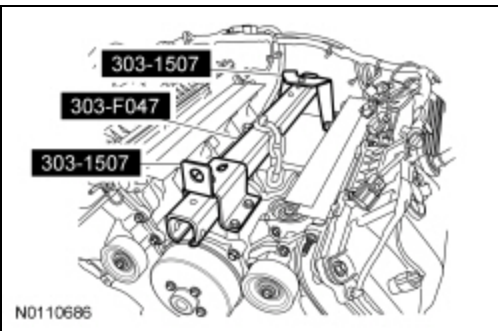
70. Connect the LH VCT variable force solenoid electrical connector and the 2 wiring harness retainers.



71. Connect the Engine Oil Pressure (EOP) switch electrical connector.



72. Install the Engine Lifting Bracket and Engine Support Brackets.

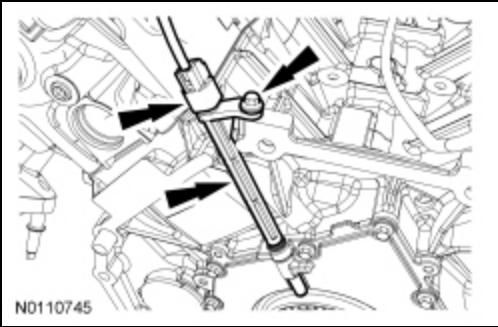


73. Using a floor crane, remove the engine from the engine stand.

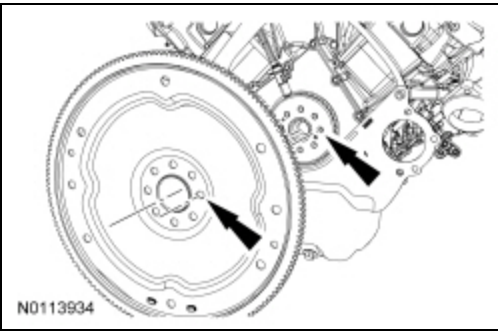
74. **NOTICE: The Crankshaft Position (CKP) sensor must be positioned into the fitting on the crankshaft rear seal retainer plate and be flush against the boss on the engine block before the bolt is installed. If the CKP sensor is installed incorrectly, the CKP sensor can be damaged.**

Position the Crankshaft Position (CKP) sensor, install the bolt and connect the electrical connector.

- Tighten to 10 Nm (89 lb-in).

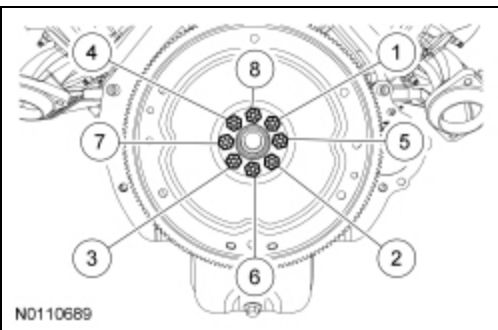


75. With the offset hole on the crankshaft ignition pulse ring aligned with the offset hole on the flexplate, position the flexplate and install 8 new bolts finger-tight.

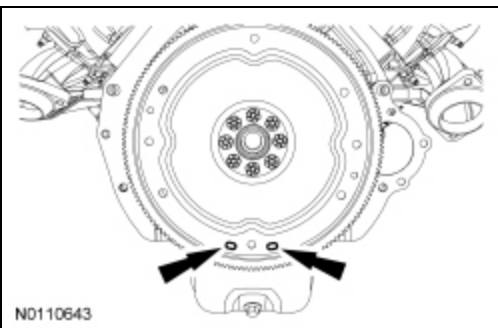


76. Tighten the 8 new bolts in 6 stages.

- Stage 1: Tighten fasteners 1 through 4 in sequence to 20 Nm (177 lb-in).
- Stage 2: Tighten fasteners 1 through 4 in sequence to 35 Nm (26 lb-in).
- Stage 3: Tighten fasteners 1 through 4 sequence an additional 60 degrees.
- Stage 4: Tighten fasteners 5 through 8 in sequence to 20 Nm (177 lb-in).
- Stage 5: Tighten fasteners 5 through 8 in sequence to 35 Nm (26 lb-in).
- Stage 6: Tighten fasteners 5 through 8 in sequence an additional 60 degrees.



77. Rotate the engine clockwise until the 2 slotted holes on the flexplate are at the 6 o'clock position as shown prior to engine installation.



78. Install the engine. For additional information, refer to Engine in this section.